$$CH_3CHO + 2Cu^{2+} + 5OH^- \longrightarrow CH_3COO^- + Cu_2O\downarrow + 3H_2O$$

(blue) (red)

iii) Benedict's solution Test:

Benedicts solution is a mixture of $CuSO_4$ + sodium citrate + NaOH. Cu^{2+} is reduced by aldehyde to give red precipitate of cuprous oxide.

$$CH_3CHO + 2Cu^{2+} + 5OH^- \longrightarrow CH_3COO^- + Cu_2O\downarrow + 3H_2O$$

(blue) (red)

iv) Schiffs' reagent Test

Dilute solution of aldehydes when added to schiffs' reagent (Rosaniline hydrochloride dissolved in water and its red colour decolourised by passing SO_2) yields its red colour. This is known as **Schiffs' test** for aldehydes . Ketones do not give this test. Acetone however gives a positive test but slowly.

12.7 Uses of Aldehydes and Ketones

Formaldehyde

(i) 40% aqueous solution of formaldehyde is called **formalin**. It is used for preserving biological specimens.

(ii) Formalin has hardening effect, hence it is used for tanning.

(iii) Formalin is used in the production of thermo setting plastic known as bakelite, which is obtained by heating phenol with formalin.

Acetaldehye

(i) Acetaldehyde is used for silvering of mirrors

(ii) Paraldehyde is used in medicine as a hypnotic.

(iii) Acetaldehyde is used in the commercial preparation of number of organic compounds like acetic acid, ethyl acetate etc.,

Acetone

(i) Acetone is used as a solvent, in the manufacture of smokeless gun powder (cordite)

(ii) It is used as a nail polish remover.

(iii) It is used in the preparation of sulphonal, a hypnotic.

(iv) It is used in the manufacture of thermosoftening plastic **Perspex.**

Benzaldehyde is used

(i) as a flavoring agent (ii) in perfumes (iii) in dye intermediates

(iv) as starting material for the synthesis of several other organic compounds like cinnamaldehyde, cinnamic acid, benzoyl chloride etc.

Aromatic Ketones

(i) Acetophenone has been used in perfumery and as a hypnotic under the name **hypnone.**

(ii) Benzophenone is used in perfumery and in the preparation of **benzhydrol eye drop.**

XII U12-Carbonyl Compounds.indd 167 11-01-2021 16:10:40